

Not due for submission, will not be marked.

The first ten questions are repeats/twists of basic material. They should provide useful exercise or practice material for exam preparation.

The last questions require more serious calculations/understanding.

Symbols may have different meanings in different questions; do not assume continuation.

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1. The set  $\{|\phi_1\rangle, |\phi_2\rangle\}$  is an orthonormal basis set for a 2-dimensional Hilbert space. The states  $|W\rangle$ ,  $|\theta_1\rangle$  and  $|\theta_2\rangle$  are defined as

$$|W\rangle = \alpha |\phi_1\rangle + i\alpha |\phi_2\rangle ,$$

$$|\theta_1\rangle = \frac{i}{\sqrt{3}} |\phi_1\rangle - \frac{\sqrt{2}}{\sqrt{3}} |\phi_2\rangle , \quad |\theta_2\rangle = |\phi_1\rangle .$$

- (a) Find the constant  $\alpha$  such that  $|W\rangle$  is normalized. Take care that you don't assume  $\alpha$  to be real or imaginary; keep your answer general.
- (b) Calculate the inner product  $\langle\theta_1|\theta_2\rangle$ .
- (c) Is the set  $\{|\theta_1\rangle, |\theta_2\rangle\}$  an orthonormal set? Explain why, or why not.
- (d) Using the representation

$$|\phi_1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} , \quad |\phi_2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} ,$$

express  $|\theta_1\rangle$  and  $|\theta_2\rangle$  as column vectors. Also write down  $\langle\theta_1|$  and  $\langle\theta_2|$  as row vectors.

- (e) Using the same representation, express the operator  $\hat{M} = |\theta_1\rangle\langle\theta_2|$  as a matrix.  
Determine whether or not  $\hat{M}$  is hermitian. Make sure you learn to tell whether a matrix is hermitian simply by looking at it.  
Also determine whether or not  $\hat{M}$  is unitary.
- (f) Calculate the expectation value of  $\hat{B} = |\phi_2\rangle\langle\phi_2|$  in the state  $|\theta_1\rangle$ .  
Calculate the uncertainty of the observable  $B$  (represented by operator  $\hat{B}$ ) in the state  $|\theta_1\rangle$ .
- (g) Calculate the expectation value and the uncertainty of  $B$  (defined above) in the state  $|\theta_2\rangle$ .

2. (a) The function  $\phi_1(x)$  is found to be an eigenstate of the Hamiltonian for a single particle in one dimension, with the corresponding energy being  $\epsilon_1$ . Show that

$$f(x, t) = \phi_1(x)e^{-i\epsilon_1 t/\hbar}$$

is a solution of the corresponding time-dependent Schroedinger equation.

- (b) Another eigenstate  $\phi_2(x)$  corresponds to eigenenergy  $\epsilon_2$ . Show that

$$g(x, t) = \phi_1(x)e^{-i\epsilon_1 t/\hbar} + \phi_2(x)e^{-i\epsilon_2 t/\hbar}$$

is a solution of the corresponding TDSE.

3. Complex numbers and functions.

- (a) Assuming  $k_1, k_2$  to be real, explain the difference between  $|k_1 + ik_2|^2$  and  $(k_1 + ik_2)^2$ . Is there ever a situation where these two are equal? Is there any situation for which you could write  $|ik_2|^2 = -k_2^2$ ?
- (b) Express  $e^{ikx}$  in terms of  $\cos(kx)$  and  $\sin(kx)$ .  
Hint: in case this is not familiar, please look up “complex exponential” or “Eulers formula”.
- (c) Express  $\cos(kx)$  in terms of  $e^{ikx}$  and  $e^{-ikx}$ .
- (d) If  $k$  is a real positive constant, is  $e^{ikx}$  a periodic function of  $x$ ? If so, what is the period?
- (e) Assuming  $k$  to be a real positive constant, is  $e^{ikx}$  an odd function of  $x$ ? Is it an even function of  $x$ ?
- (f) If  $\lambda$  is a real number, what is the norm/magnitude of  $e^{i\lambda}$ ?
- (g) If  $\lambda = \lambda_1 + i\lambda_2$  is a complex number with real part  $\lambda_1$  and imaginary part  $\lambda_2$ , what is the norm/magnitude of  $e^{i\lambda}$ ?
- (h) If  $|\alpha|^2 = 4$ , can I infer that  $\alpha = 2$  necessarily? Introduce a phase factor to write the general solution for  $\alpha$ . What is a ‘phase factor’? Draw the curve in the complex  $\alpha$  plane which represents all possible solutions for  $\alpha$ .

4. Consider a particle of mass  $m$  on a one-dimensional line.
- (a) Write down the operators for momentum and for kinetic energy.
  - (b) Is the function  $e^{ikx}$  an eigenfunction of the momentum operator? If so, what is the eigenvalue?  
Answer the same questions for the functions  $-e^{ikx}$ ,  $C_1 e^{ikx}$ ,  $C_2 e^{-ikx}$ , where  $C_1$  and  $C_2$  are real or complex constants.  
Answer the same questions for the functions  $e^{ikx} + e^{-ikx}$  and  $C_1 e^{ikx} + C_2 e^{-ikx}$ . What goes wrong in these cases?
  - (c) Show that the functions  $\cos(kx)$  and  $\sin(kx)$  are not eigenfunctions of the momentum operator.  
Find a linear combination of  $\cos(kx)$  and  $\sin(kx)$  which is an eigenfunction of the momentum operator.
  - (d) Find out whether the functions  $\cos(kx)$ ,  $\sin(kx)$ ,  $e^{ikx}$ ,  $e^{-ikx}$  are eigenfunctions of the kinetic energy operator. If so, find the corresponding eigenvalues.
5. An operator  $\hat{A}$  acts on the space of functions of a real variable, i.e., on the space of possible wavefunctions of a single particle in one dimension.
- (a) If  $f(x)$  is an eigenfunction of  $\hat{A}$ , is  $-f(x)$  also an eigenfunction? If so, does it correspond to the same eigenvalue?
  - (b) If  $f(x)$  is an eigenfunction of  $\hat{A}$ , is  $Cf(x)$  also an eigenfunction? Here  $C$  is a complex constant. If  $Cf(x)$  is an eigenfunction as well, does it correspond to the same eigenvalue?
  - (c) If  $f(x)$  and  $g(x)$  are eigenfunctions of  $\hat{A}$  corresponding to the *same* eigenvalue, show that any linear combination  $C_1 f(x) + C_2 g(x)$  is also an eigenfunction of  $\hat{A}$ .
  - (d) If  $f(x)$  and  $g(x)$  are eigenfunctions of  $\hat{A}$  corresponding to *different* eigenvalues, find out whether any linear combination  $C_1 f(x) + C_2 g(x)$  is also an eigenfunction of  $\hat{A}$ .

6. Consider a single particle in one dimension, subject to a constant (space-independent) potential  $V(x) = V_0$ .
- (a) Write down the time-independent Schroedinger equation.
  - (b) Show that  $\phi_1(x) = e^{ikx}$  is a solution to this equation. Determine the energy  $E$  corresponding to this equation, in terms of  $k$  and  $V_0$ .
  - (c) Is the function  $\phi_2(x) = C\phi_1(x) = Ce^{ikx}$ , where  $C$  is a constant, also a solution?
  - (d) What is a solution of the form  $e^{ikx}$  called? Is it right-moving or left-moving? Explain why.
  - (e) Show that  $\phi_3(x) = \cos(kx)$  is a solution to the time-independent Schroedinger equation. Write down the corresponding solution to the time-dependent Schroedinger equation.
  - (f) Show that  $\phi_4(x) = C_1 \cos(kx) + C_2 \sin(kx)$  is also an eigenvalue of the Hamiltonian, where  $C_1$  and  $C_2$  are any complex constants.
  - (g) Find out if  $\phi_5(x) = C_1 \cos(k_1x) + C_2 \sin(k_2x)$  is an eigenvalue, if  $k_1 \neq k_2$ .
  - (h) I have used above the phrases “solution to the time-independent Schroedinger equation” and “eigenvalue of the Hamiltonian”. Explain why these mean the same thing.
7. Consider a single particle in one dimension, subject to a constant potential  $V(x) = V_0$  in some region.
- (a) Write down the time-independent Schroedinger equation for this region.
  - (b) Assuming  $E > V_0$ , write down the general solution for an energy eigenstate. If you introduce constants, relate them to  $E$  and  $V_0$  if possible.
  - (c) Write down the general solution for an energy eigenstate corresponding to  $E < V_0$ . Again, if any constants are related to  $E$  and  $V_0$ , write down the relationships explicitly.

8. Consider a system with a 4-dimensional Hilbert space. We are given the set

$$|w_1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad |w_2\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad |w_3\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad |w_4\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix},$$

- (a) Is the set of four vectors  $\{|w_i\rangle\}$  orthonormal?  
 (b) Does this set span the Hilbert space?  
 (c) Find the expectation value and uncertainty of the physical quantity represented by the operator  $\hat{P}_1 = |w_2\rangle\langle w_2|$  in the state

$$|\theta_3\rangle = -\frac{i}{2}|w_1\rangle - \frac{1}{2}|w_1\rangle + \frac{i}{2}|w_1\rangle + \frac{1}{2}|w_4\rangle$$

Do this in two different ways: first, keeping all operators and states in terms of  $|w_i\rangle$ 's and  $\langle w_j|$ 's, and second, expressing operators/states as matrices/vectors.

Which do you find less cumbersome?

- (d) In the same state  $|\theta_3\rangle$ , find the expectation value and then the uncertainty of

$$\hat{P}_2 = (|w_1\rangle + |w_2\rangle)(\langle w_1| + \langle w_2|)$$

9. The operator  $\hat{Y}$  has eigenvalues  $y_n$  and eigenvectors  $|w_n\rangle$ :

$$\hat{Y}|w_n\rangle = y_n|w_n\rangle; \quad n = 1, 2, 3, \dots, 6$$

- (a) If the corresponding observable  $Y$  is measured, what are the possible results?  
 (b) The system is prepared in the state

$$|\psi\rangle = \frac{1}{2}|w_1\rangle + \frac{i}{2}|w_3\rangle - \frac{i}{\sqrt{2}}|w_5\rangle$$

and a measurement of  $Y$  is performed. What are the possible results, and with what probability do they occur?

- (c) What is the dimension of the Hilbert space?

10. Consider a single particle in the one-dimensional harmonic potential  $V(x) = \frac{1}{2}m\omega^2x^2$ . Consider the wavefunction  $\phi(x) = Axe^{-x^2/2\sigma^2}$ .
- (a) Write down the time-independent Schroedinger equation.
  - (b) Show that the wavefunction  $\phi(x)$  is a solution of this equation, i.e., is an eigenfunction of the Hamiltonian, only if  $\sigma$  has a certain value. Find this value of  $\sigma$ .
  - (c) What is the energy corresponding to this eigenfunction?
  - (d) Based on the number of nodes in the wavefunction, argue whether this is the ground state, the first excited state, or the second excited state.
11. (a) Show that the hermitian conjugate of  $\hat{A}\hat{B}$  is  $\hat{B}^\dagger\hat{A}^\dagger$ .
- (b) If  $\hat{A}$  is hermitian, show that so is  $\hat{A}^2$ .
  - (c) If  $\hat{M}$  is hermitian and  $\alpha$  is a real number, show that  $\hat{P} = e^{\alpha\hat{M}}$  is also hermitian. Recall that the exponential of an operator is defined in terms of a power series.
  - (d) If  $\hat{M}$  is hermitian and  $\alpha$  is a real number, show that  $\hat{Q} = e^{i\alpha\hat{M}}$  is unitary.
12. The generalized uncertainty relation states that, if two observables  $A$  and  $B$  are described by the operators  $\hat{A}$  and  $\hat{B}$ , then

$$\Delta A \Delta B \geq \left| \frac{i}{2} \langle [\hat{A}, \hat{B}] \rangle \right| \quad \text{where} \quad \begin{cases} \Delta A = \sqrt{\langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2} \\ \text{and similarly for } \Delta B \end{cases}$$

with all expectation values taken in the same state.

Let's apply this to the case of one particle in one dimension.

- (a) Apply this to the case where the observables are position and momentum. You should obtain the Heisenberg uncertainty relation.
- (b) Apply this to the case where the observables are position and kinetic energy. You may need to calculate the commutation relation between position and kinetic energy.

- (c) Apply this to the case where the observables are momentum and kinetic energy.

13. Consider two particles confined to a single line (one dimension), so that the wavefunctions of the system are of the form  $\psi(x_A, x_B, t)$ , where  $x_A$  and  $x_B$  are the coordinates of the two particles. The time-independent Schroedinger equation

$$\hat{H}\psi(x_A, x_B) = E\psi(x_A, x_B)$$

is found to have eigenstates  $w_i(x_A, x_B)$  and corresponding eigenvalues  $E_i$ , with  $i = 1, 2, 3, \dots$ . Show that

$$f(x, t) = C_1 w_1(x_A, x_B) e^{-iE_1 t/\hbar} + C_2 w_2(x_A, x_B) e^{-iE_2 t/\hbar}$$

is a solution of the corresponding time-dependent Schroedinger equation.

Hint: it might make sense to write down the time-dependent Schroedinger equation first.

14. Consider the wavefunction of a single particle

$$\phi(x) = N e^{-|x|/\lambda} = \begin{cases} N e^{-x/\lambda} & \text{for } x > 0 \\ N e^{+x/\lambda} & \text{for } x < 0 \end{cases}$$

- Calculate the constant  $N$  so that the wavefunction is normalized.
- Calculate the expectation value of the position in this state. (Guess the value before performing the calculation.)
- Calculate the uncertainty of the position in this state. Before performing the calculation, guess the  $\lambda$ -dependence of the uncertainty.
- Calculate the expectation value of the momentum in this state. Guess the value before performing the calculation.
- Calculate the uncertainty of the momentum in this state. Before performing the calculation, guess the  $\lambda$ -dependence of the uncertainty.
- You should recognize the wavefunction as the ground state of a particular potential. Which potential?

15. Consider the wavefunction of the two-state system

$$|\psi(t)\rangle = \alpha(t) \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta(t) \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \alpha(t) \\ \beta(t) \end{pmatrix}.$$

If the Hamiltonian of the system is

$$\hat{H} = \begin{pmatrix} E & -\epsilon \\ -\epsilon & E \end{pmatrix}$$

use the time-dependent Schroedinger equation to find the probability amplitudes  $\alpha(t)$  and  $\beta(t)$  as functions of time. The initial conditions are  $\alpha(0) = 1$ ,  $\beta(0) = 0$ .

Comment: a system with Hilbert space of dimension two is often called a two-state system.

16. Consider a single particle in one dimension, subject to the negative delta function potential

$$V(x) = -\lambda\delta(x).$$

We considered this potential in Problem Set 8, and calculated the bound state. Now investigate the scattering properties of this potential, i.e., consider solutions with energy  $E > 0$ .

- (a) Write down plane wave solutions for the left and right half-line. Identify terms representing incident, reflected and transmitted waves.
- (b) Use the two boundary conditions, one for the continuity of the wavefunction and one for the discontinuity of its derivatives, derived in problem set 8.
- (c) Using the two conditions, find the reflection and transmission coefficients. Check that their sum makes sense.
- (d) You should be able to compare your results with that in the wikipedia page on the delta potential. The problem is probably worked out in some textbooks as well.

17. Consider a single particle of mass  $m$  in one dimension subject to no potential, i.e., a free particle. Imagine starting with the initial wavefunction

$$\psi(x, 0) = \frac{1}{(2\pi b^2)^{1/4}} \exp\left[-\frac{x^2}{4b^2}\right]$$

so that the initial probability density  $|\psi(x, 0)|^2$  is a gaussian of width  $b$  centered around the origin.

How will this wavefunction evolve? I read somewhere:

$$\psi(x, t) = \frac{1}{(2\pi b^2)^{1/4}} \frac{1}{\sqrt{1 + i(\hbar t/2mb^2)}} \exp\left[-\frac{x^2}{4b^2 [1 + i(\hbar t/2mb^2)]}\right]$$

- (a) Check that the claimed  $\psi(x, t)$  indeed is a solution of the TDSE of the free particle in one dimension. I may have made a mistake translating all these factors; if so, please correct.
- (b) Check that the wavefunction  $\psi(x, t)$  is normalized.
- (c) Show that the probability density remains of gaussian form; however it's width changes. How does its width change? Plot the width as a function of time.
- (d) What determines the speed at which the width changes (spreading speed)? Would an initially narrower wavepacket spread out faster or slower? Do you have an intuitive explanation?
- (e) Why does the wavefunction evolve at all? Did we not start with the eigenstate of an harmonic oscillator? Don't eigenstates have simple time dependence (given by a phase factor)?
- (f) Guess (explaining physically) and then calculate the expectation value of the position. How does this expectation value evolve with time?
- (g) Guess (explaining physically) and then calculate the uncertainty of the position ( $\Delta x$ ). How does this uncertainty evolve with time?
- (h) Guess the expectation value and uncertainty of the momentum, up to a constant. How do you expect these to evolve with time?  
Calculate these quantities.
- (i) Calculate the probability density and the probability current density. Use these to check whether the continuity equation is satisfied.

18. The generalized uncertainty relation states that, if two observables  $A$  and  $B$  are described by the operators  $\hat{A}$  and  $\hat{B}$ , then

$$\Delta A \Delta B \geq \left| \frac{i}{2} \langle [\hat{A}, \hat{B}] \rangle \right| \quad \text{where} \quad \begin{cases} \Delta A = \sqrt{\langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2} \\ \text{and similarly for } \Delta B \end{cases}$$

with all expectation values taken in the same state.

Let's check this for the case of a spin-1/2 system. Recall that the spin operators can be obtained from the Pauli matrices as  $\hat{S}_x = \frac{\hbar}{2}\sigma_x$ ,  $\hat{S}_y = \frac{\hbar}{2}\sigma_y$ ,  $\hat{S}_z = \frac{\hbar}{2}\sigma_z$ .

- (a) Using the commutation relation  $[\hat{S}_x, \hat{S}_y] = i\hbar\hat{S}_z$ , the generalized uncertainty principle implies that we should obtain

$$\Delta S_x \Delta S_y \geq \left| \frac{i}{2} \langle \hat{S}_z \rangle \right|$$

for any state.

Is this true? Let's check for a relatively general state:  $\psi_1 = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ , with real  $\alpha$  and  $\beta$ . This is not the most general state, since we could have complex components in general, but this will be tedious enough. Also, note that  $\alpha$  and  $\beta$  are related by normalization and hence are not independent.

By calculating  $\Delta S_x$ ,  $\Delta S_y$  and  $\langle \hat{S}_z \rangle$  for this state, show that the above relation is satisfied.

- (b) Next, show that the same uncertainty relation is satisfied in the state  $\psi_1 = \begin{pmatrix} \alpha \\ i\beta \end{pmatrix}$ , with real  $\alpha$  and  $\beta$ .
- (c) Do you find the *equality* to be satisfied in each case? Can you think up a state where the uncertainty product is actually larger than the minimum value imposed by the relation? Or is that not possible?
- (d) Do a similar check for the uncertainty product of the operators  $\hat{S}_y$  and  $\hat{S}_z$ .